

# Set Cover Problem

$A = \{1, 2, 3, 4, 5\} \rightarrow \text{set}$

$B_1 = \{4, 1, 3\} \rightarrow \underline{5 \text{ cost}}$

$B_2 \Rightarrow \{2, 5\} \Rightarrow \boxed{10 \text{ cost}}$

$B_3 \rightarrow \{1, 4, 3, 2\} \Rightarrow \boxed{3 \text{ cost}}$

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Set cover problem (combination of subsets)

$\rightarrow$  find that suitable subset that covers all elements of set with less cost.

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$$B_1 \cup B_2 = \{4, 1, 3, 2, 5\} \Rightarrow 15 \text{ cost} \checkmark$$

$$B_2 \cup B_3 = \{1, 4, 3, 2, 5\} \Rightarrow 13 \text{ cost} \checkmark$$

Does both the union cover all the elements?



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$B_1 \cup B_3 = \{1, 4, 3, 2\} \rightarrow 8 \text{ cost} \times$

Does the union ( $B_1$  and  $B_3$ ) cover all the elements?



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$$B_1 \cup B_2 \cup B_3 = \{1, 4, 3, 2, 5\} \rightarrow 18 \text{ cost} \checkmark$$

Which particular combination is covering all the elements with less cost?



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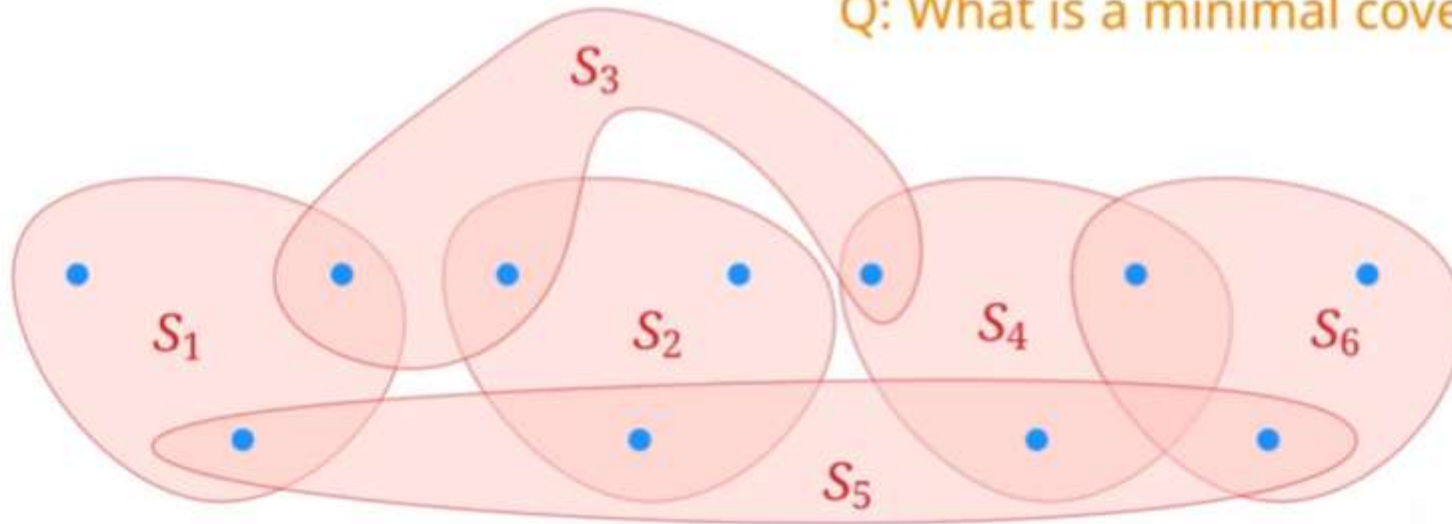
It is  $B_2 \cup B_3$

## SETCOVER (cardinality)

Let  $U$  be some **ground set** (universe),  
and let  $\mathcal{S}$  be a family of **subsets** of  $U$  with  $\bigcup \mathcal{S} = U$ .

Find a **cover**  $\mathcal{S}' \subseteq \mathcal{S}$  of  $U$  (i.e., with  $\bigcup \mathcal{S}' = U$ ) of minimum cardinality.

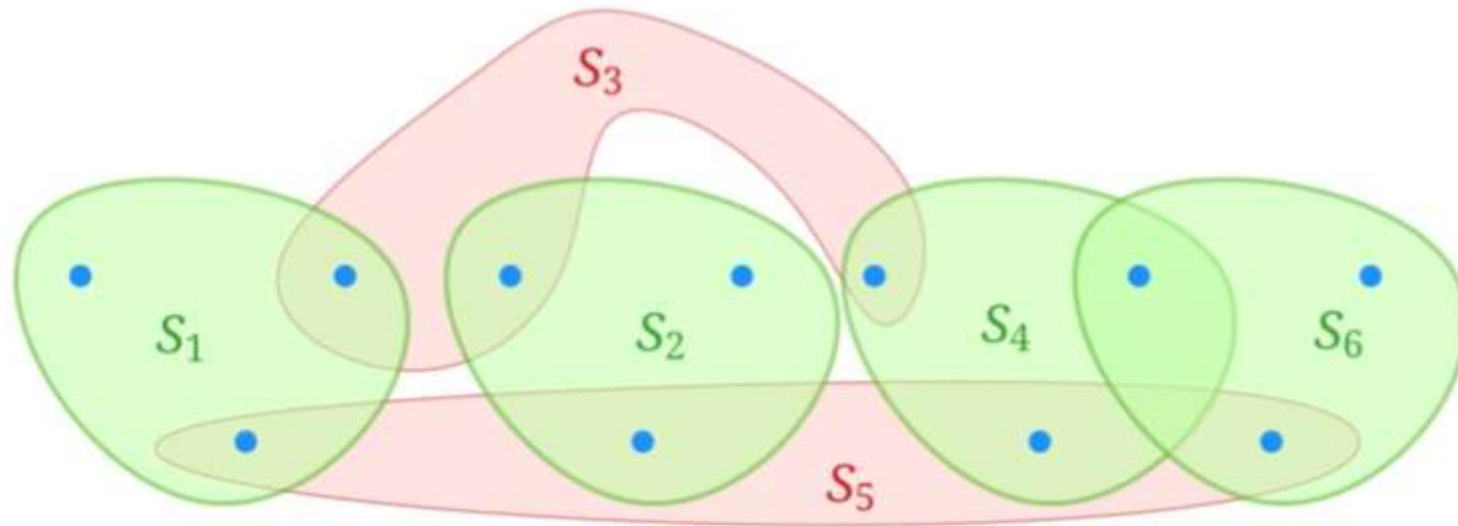
Q: What is a minimal cover here?



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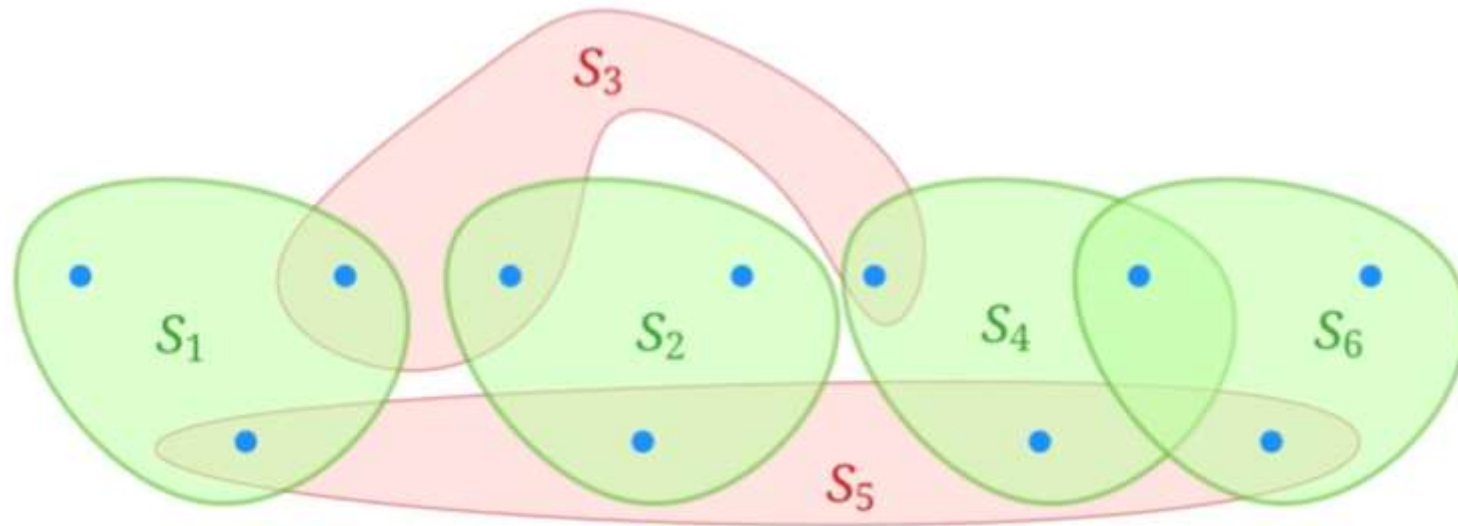


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Each  $S \in \mathcal{S}$  has cost  $c(S) > 0$ .

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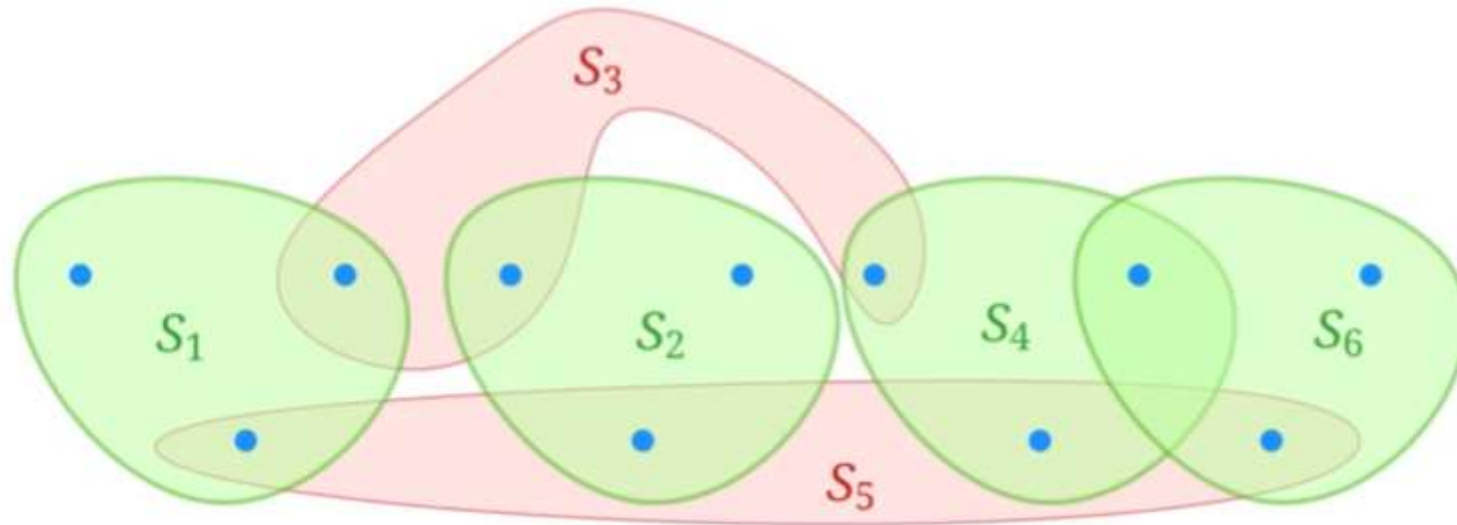


## SETCOVER (general)

Let  $U$  be some ground set (universe),  
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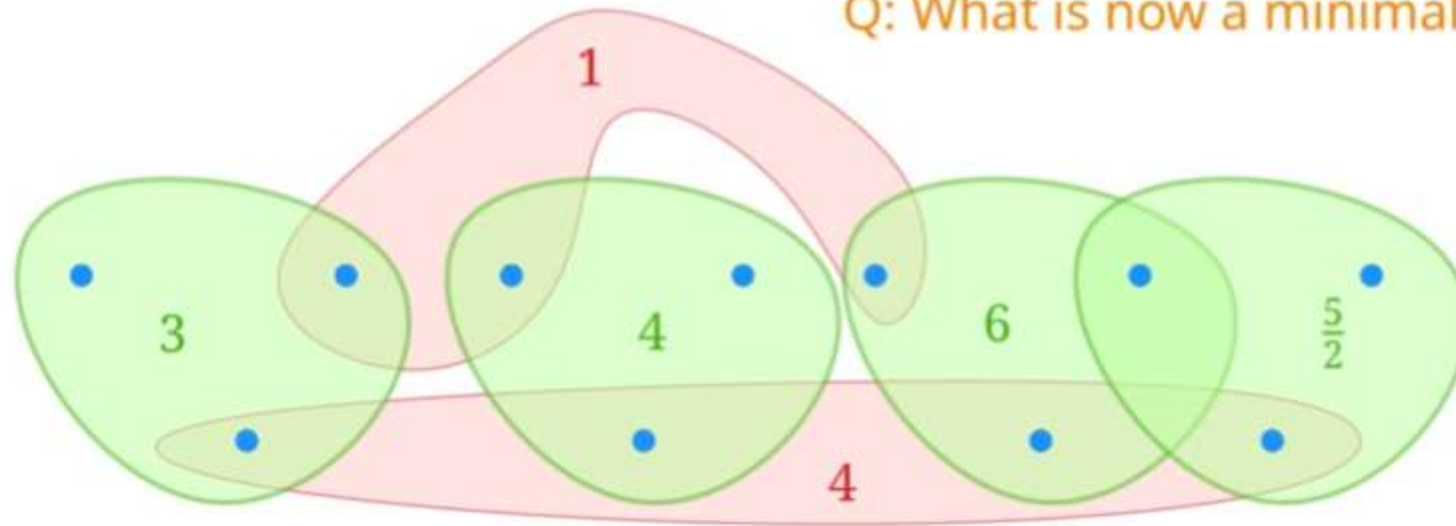
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Q: What is now a minimal cover?





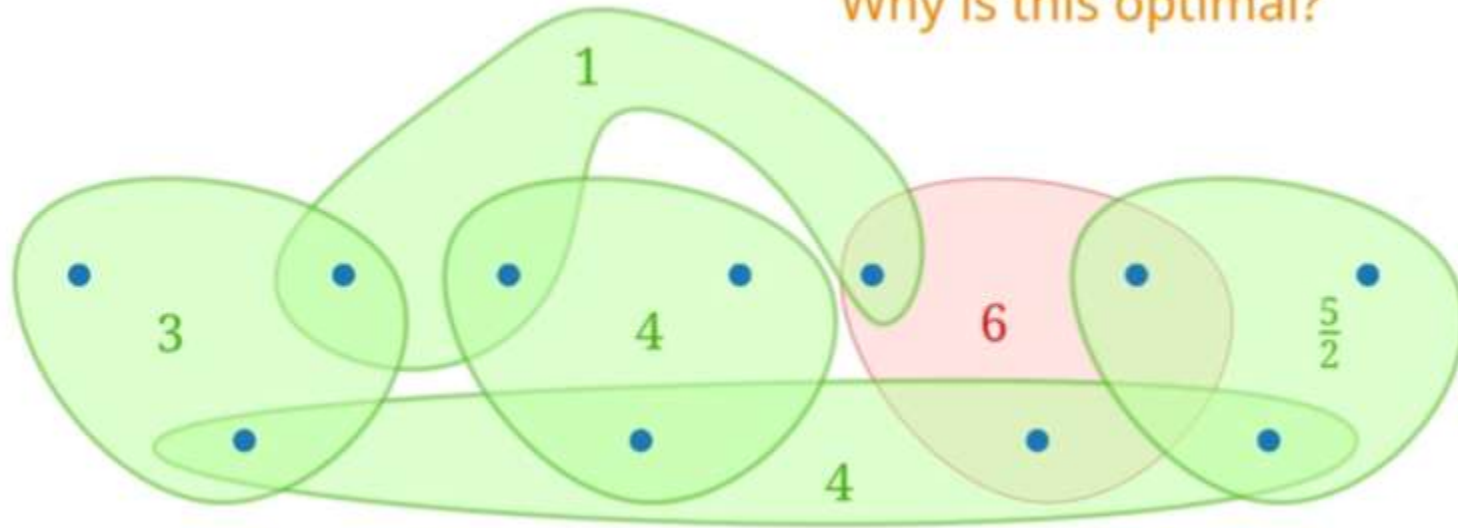
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Why is this optimal?

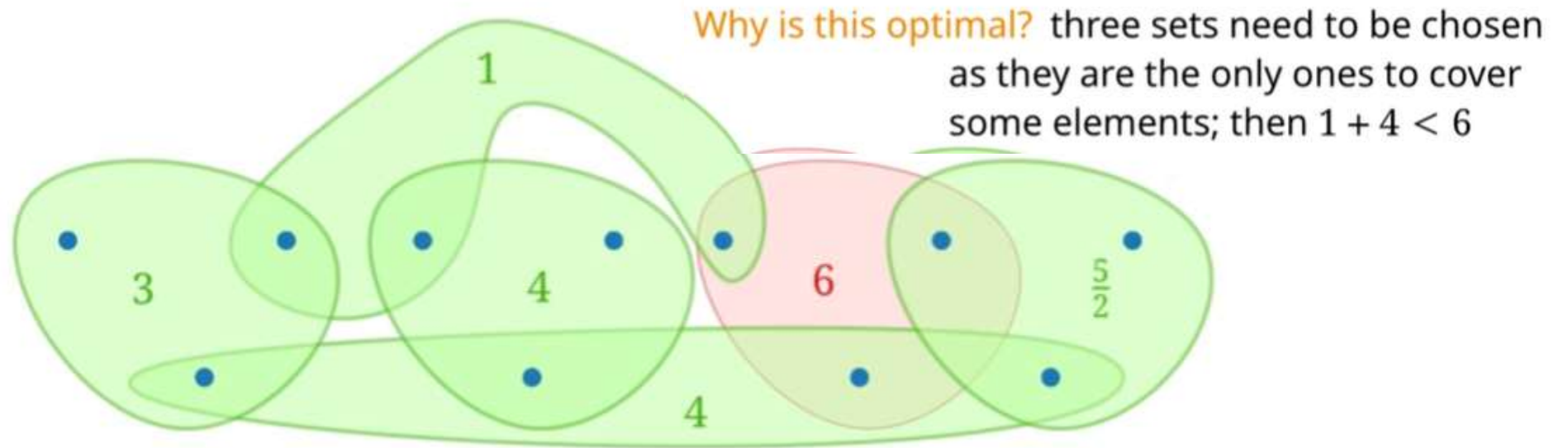


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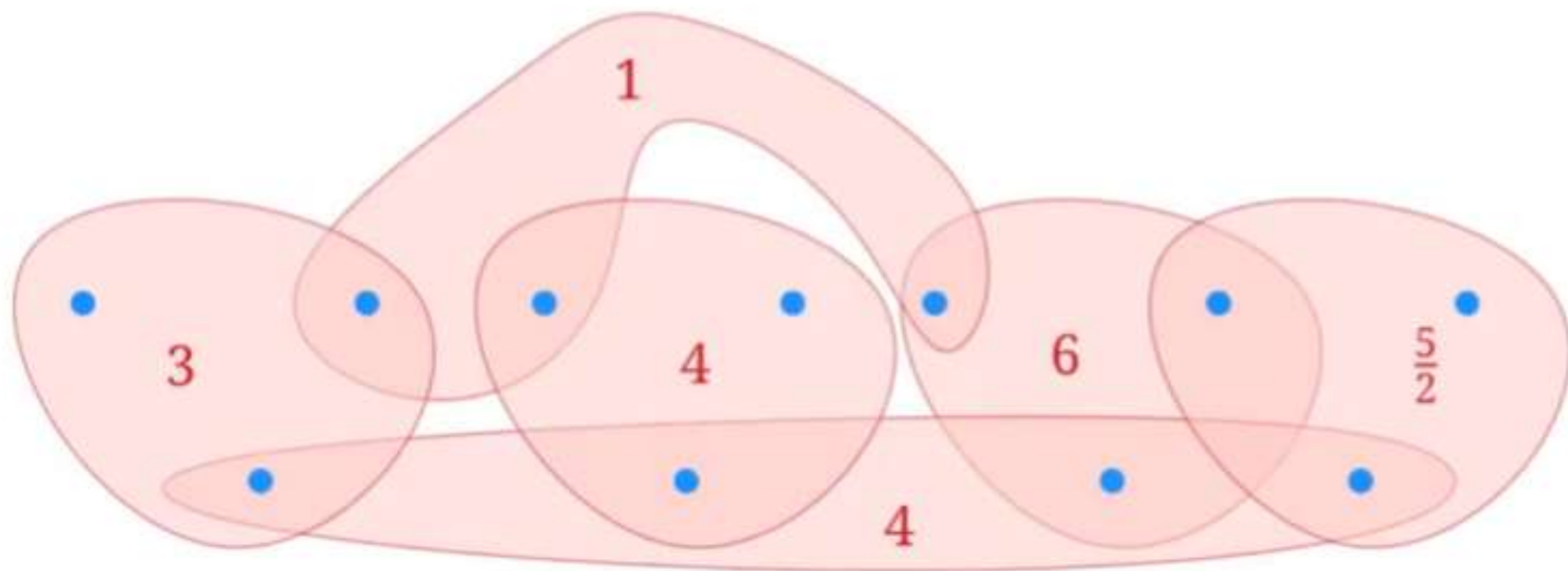


## Greedy Approximation Algorithm for SETCOVER



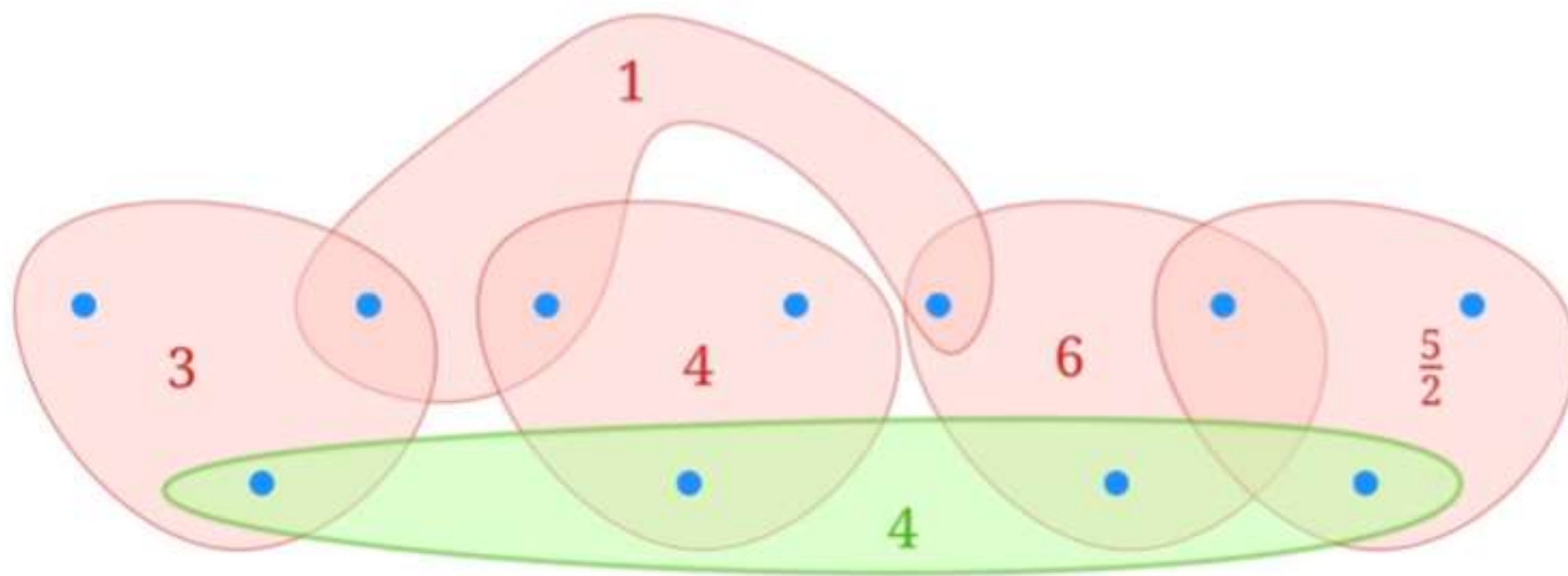
# Iterative “Buying” of Elements

What is the real cost of picking a set?



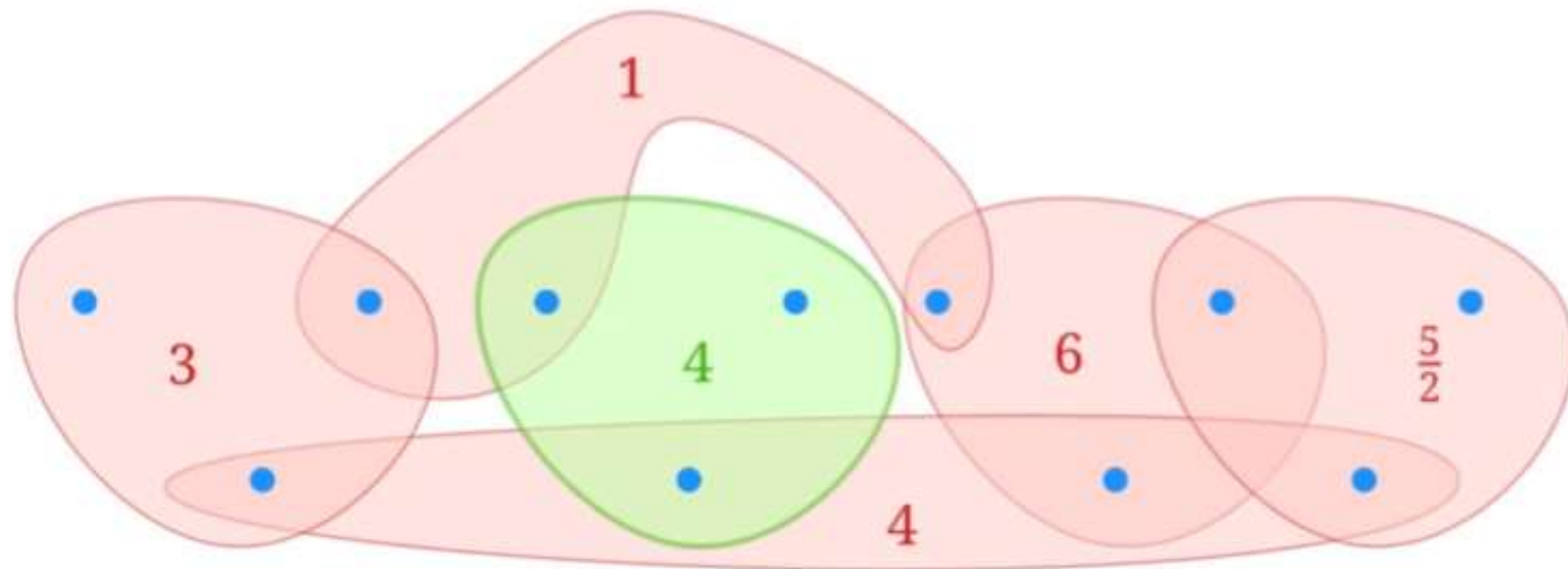
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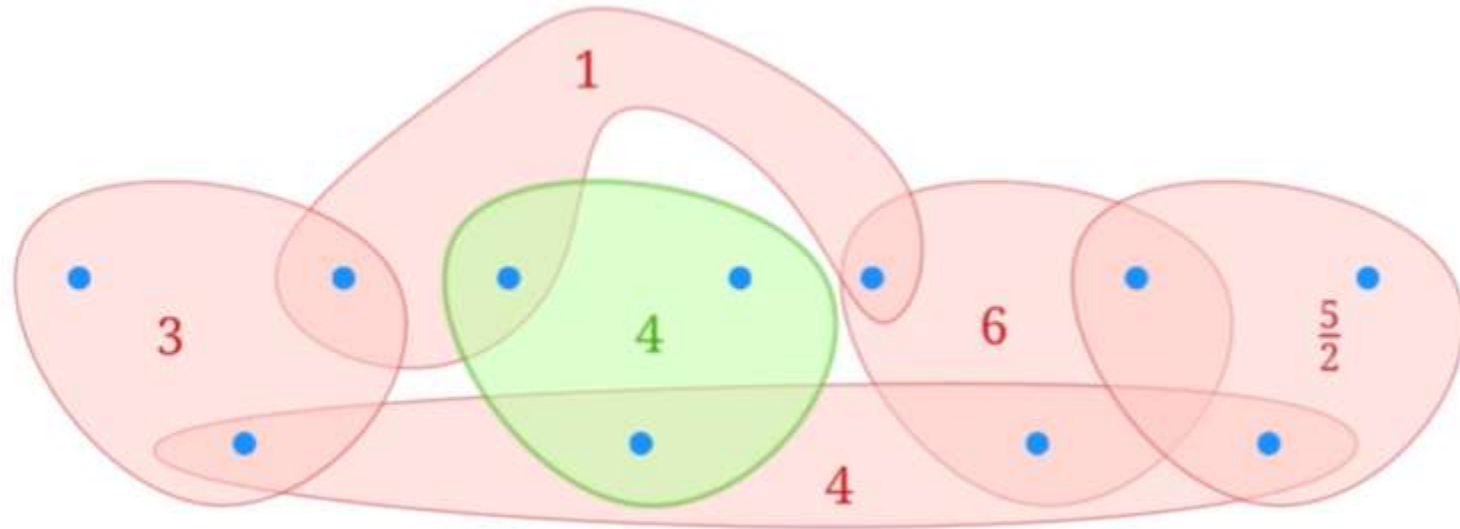
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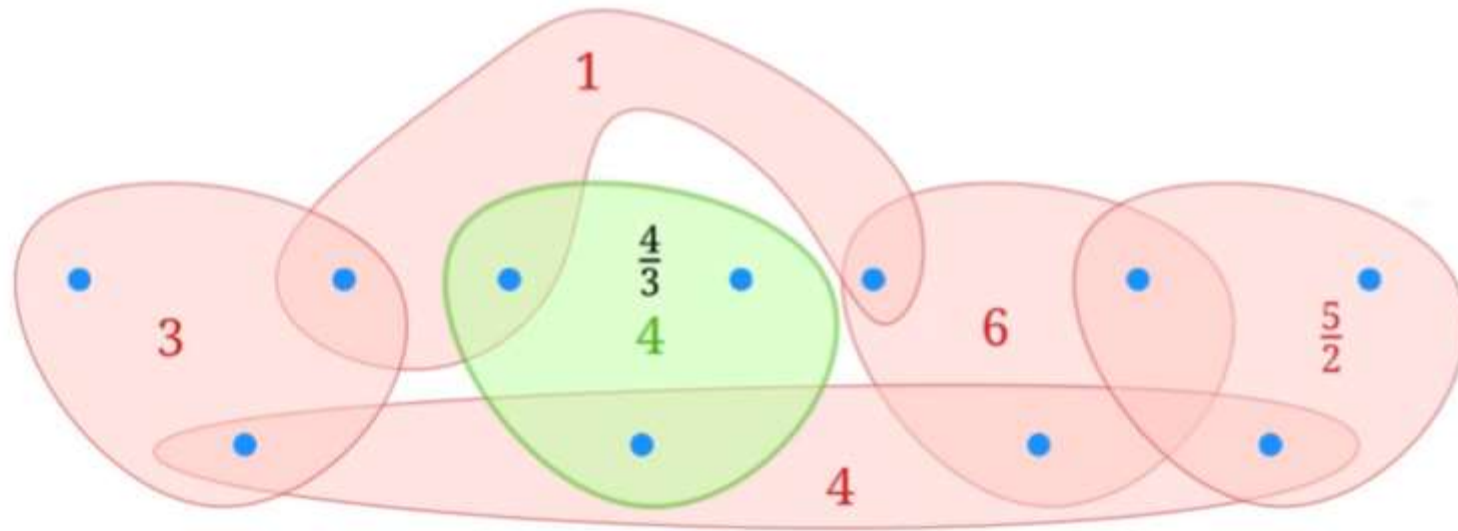




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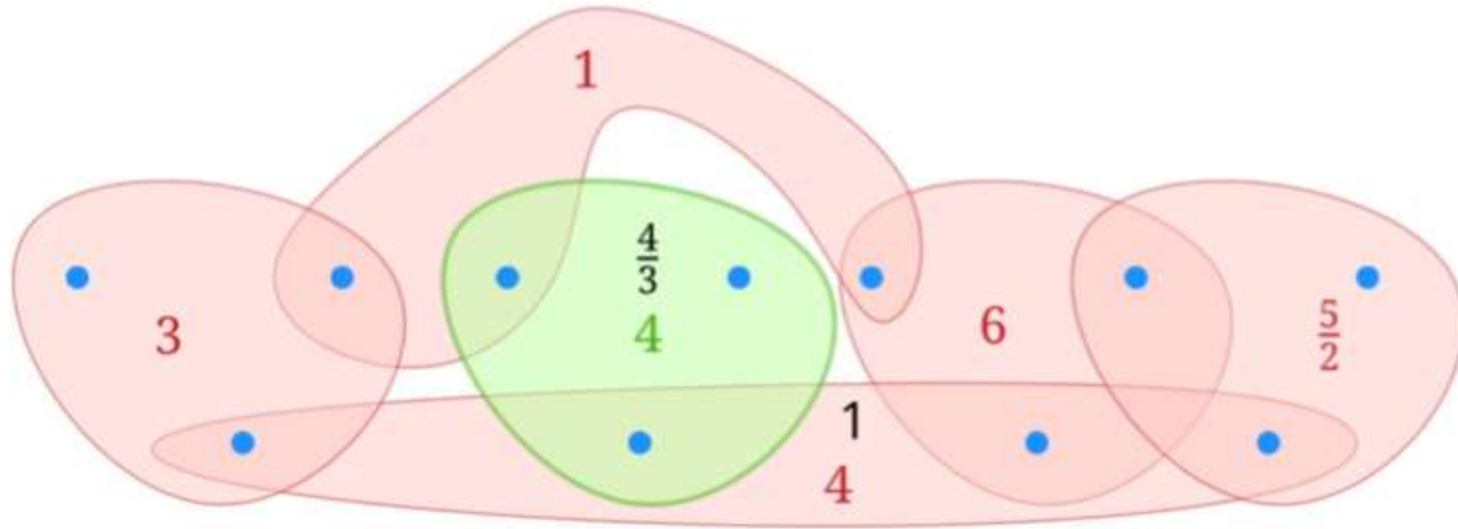
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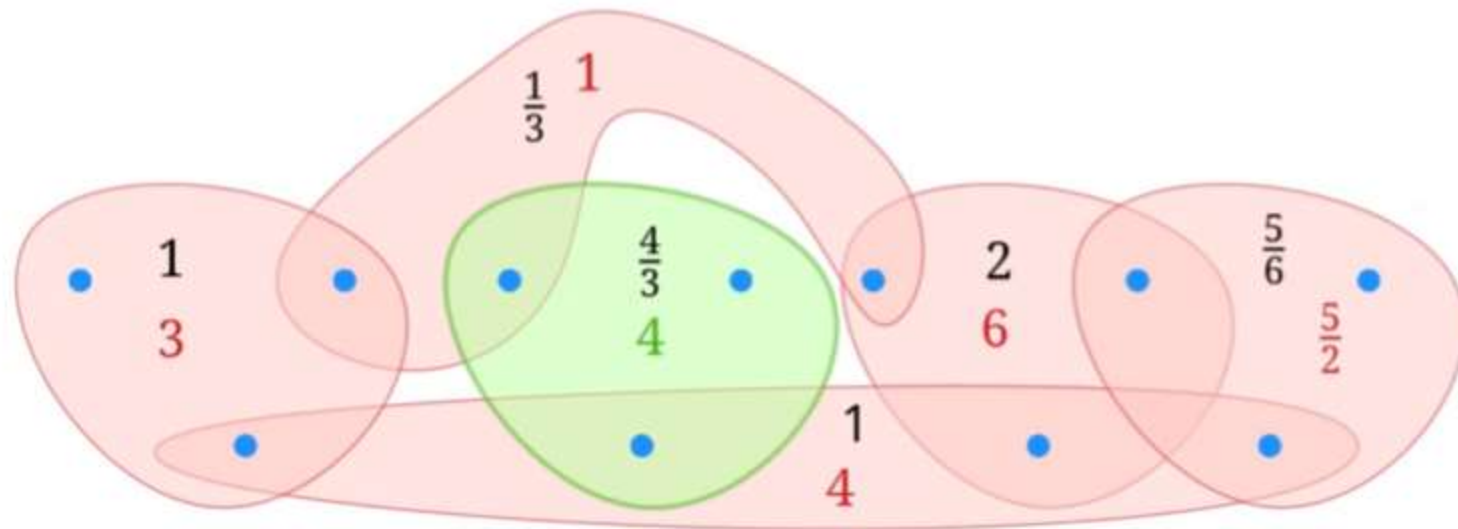
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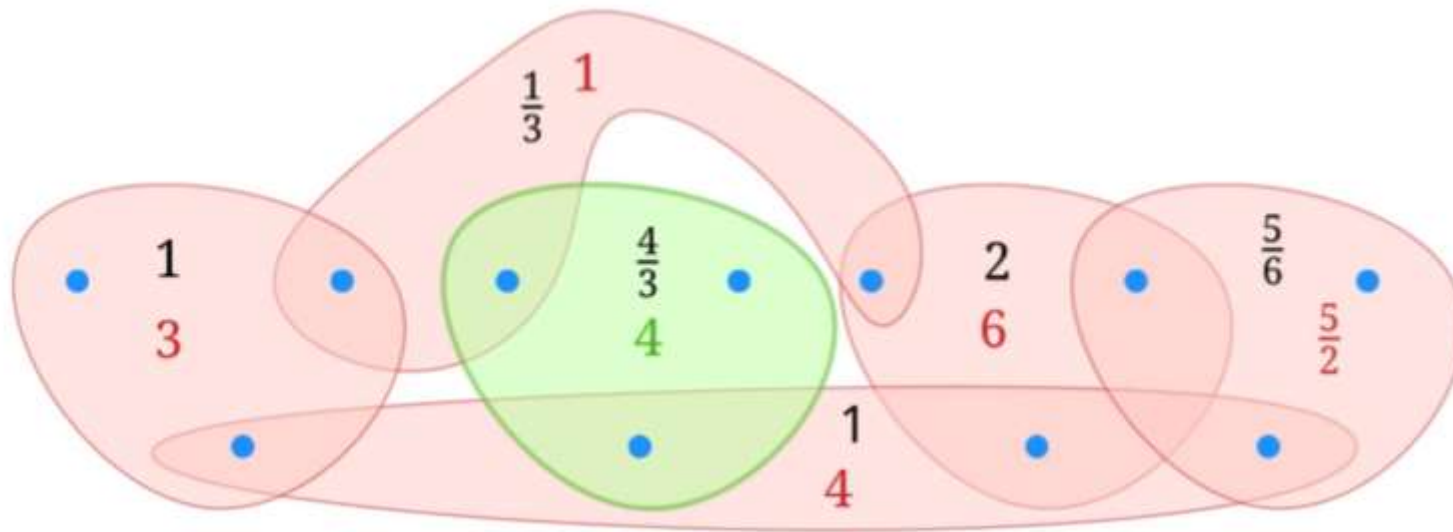


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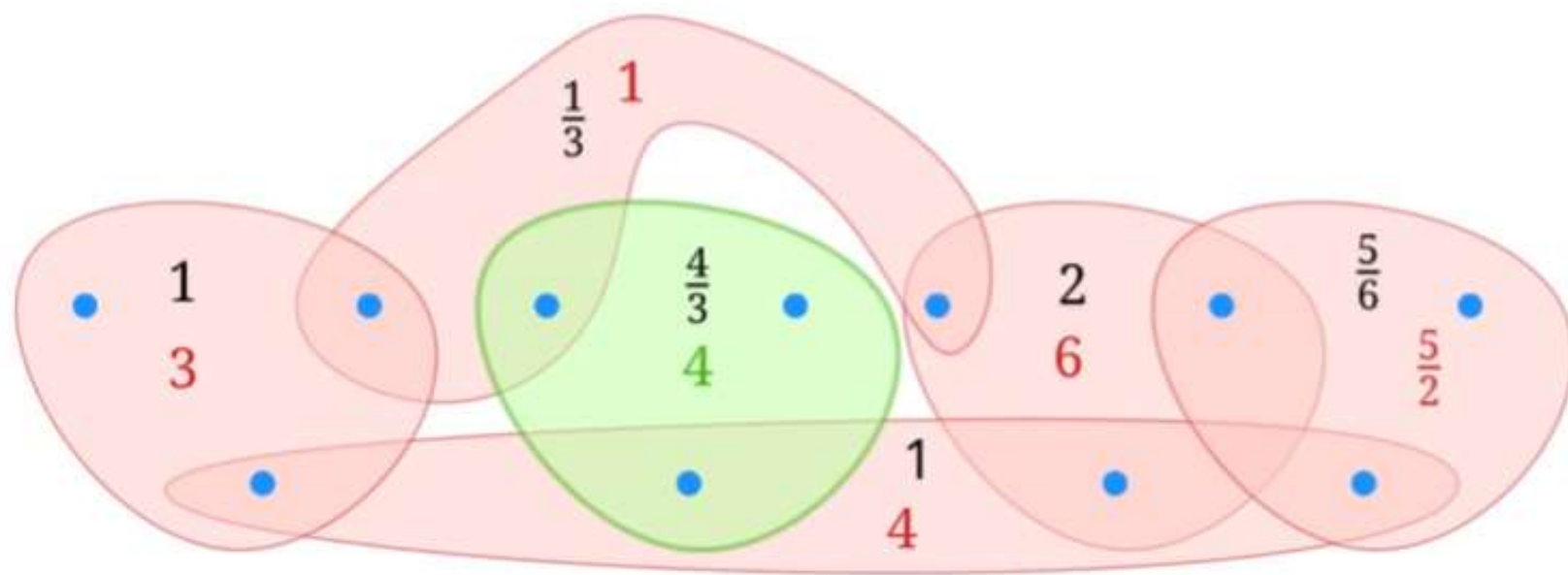
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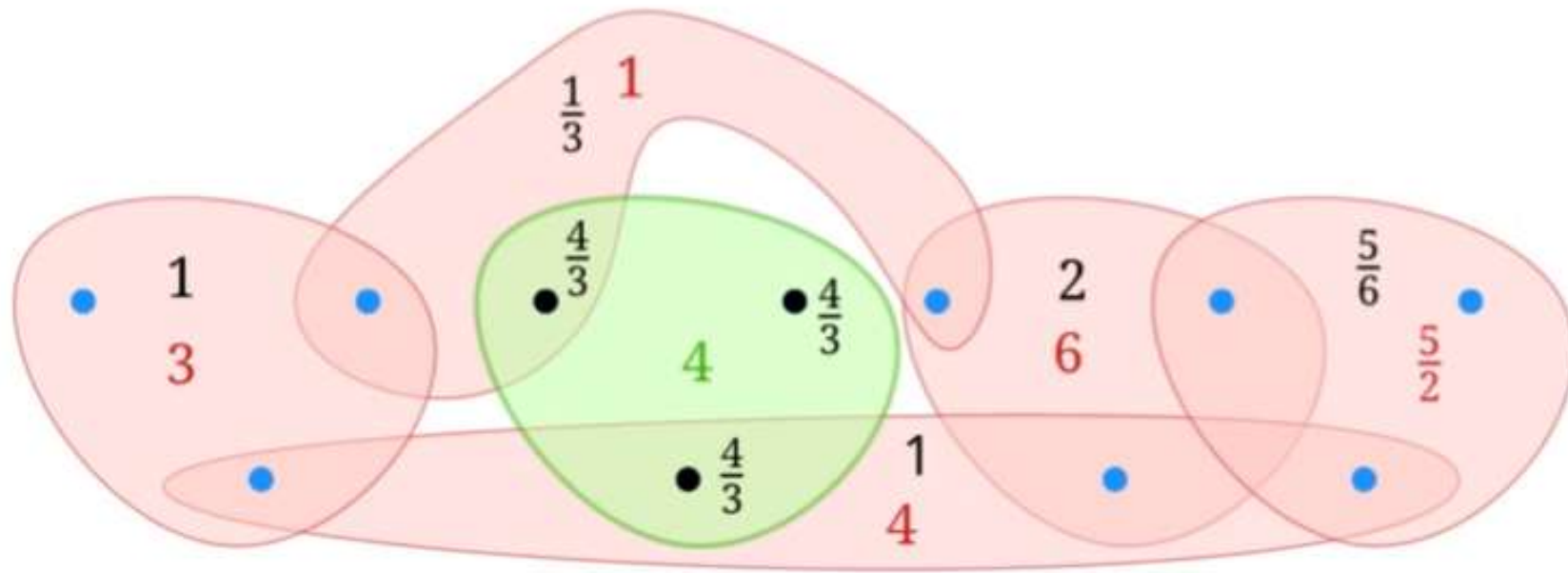
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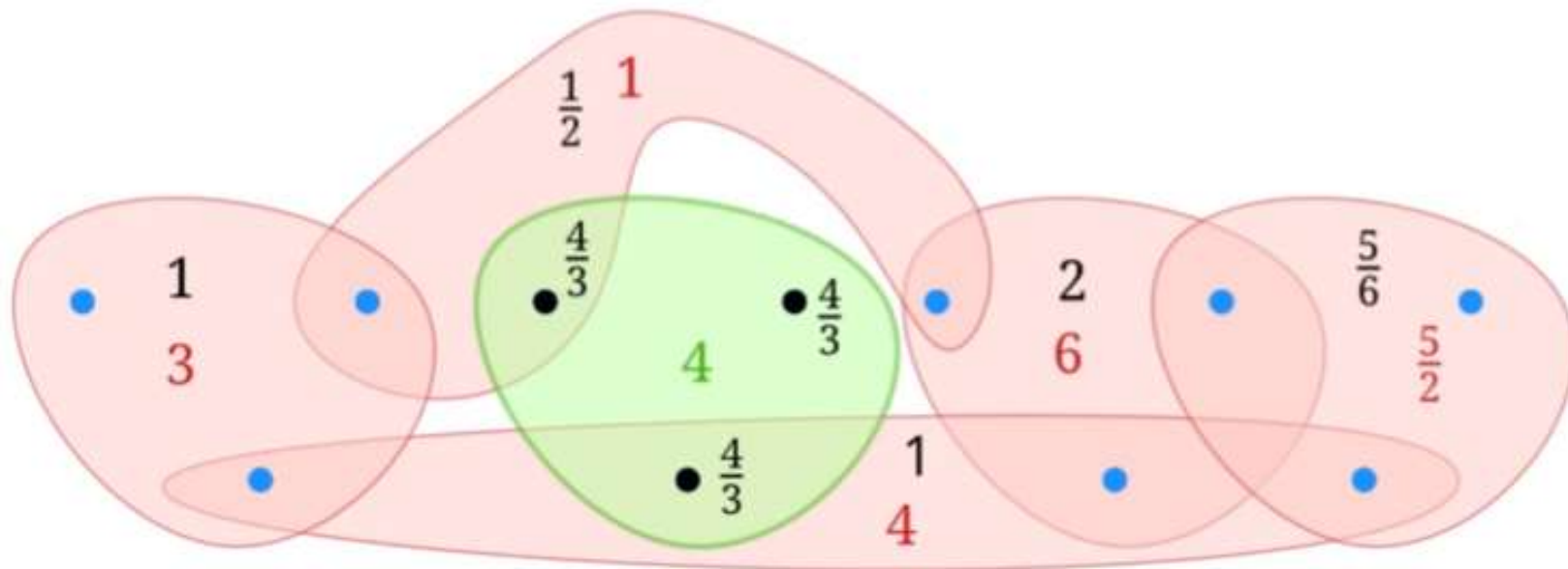
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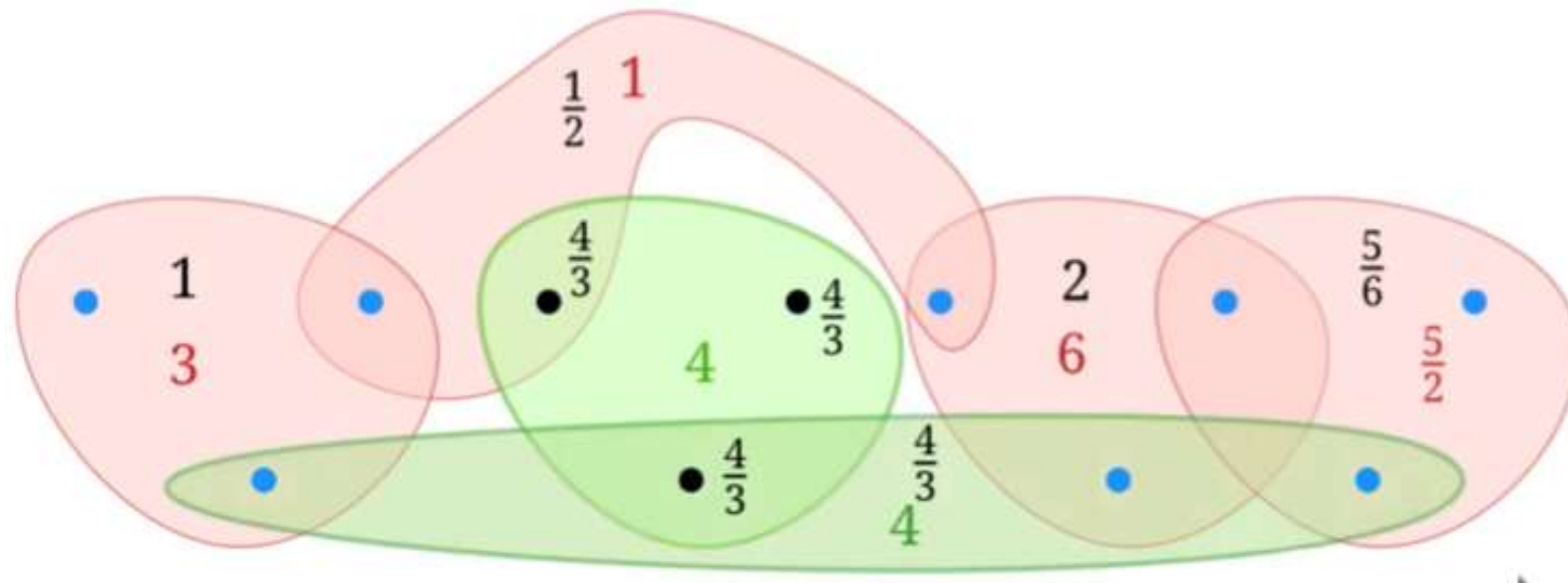
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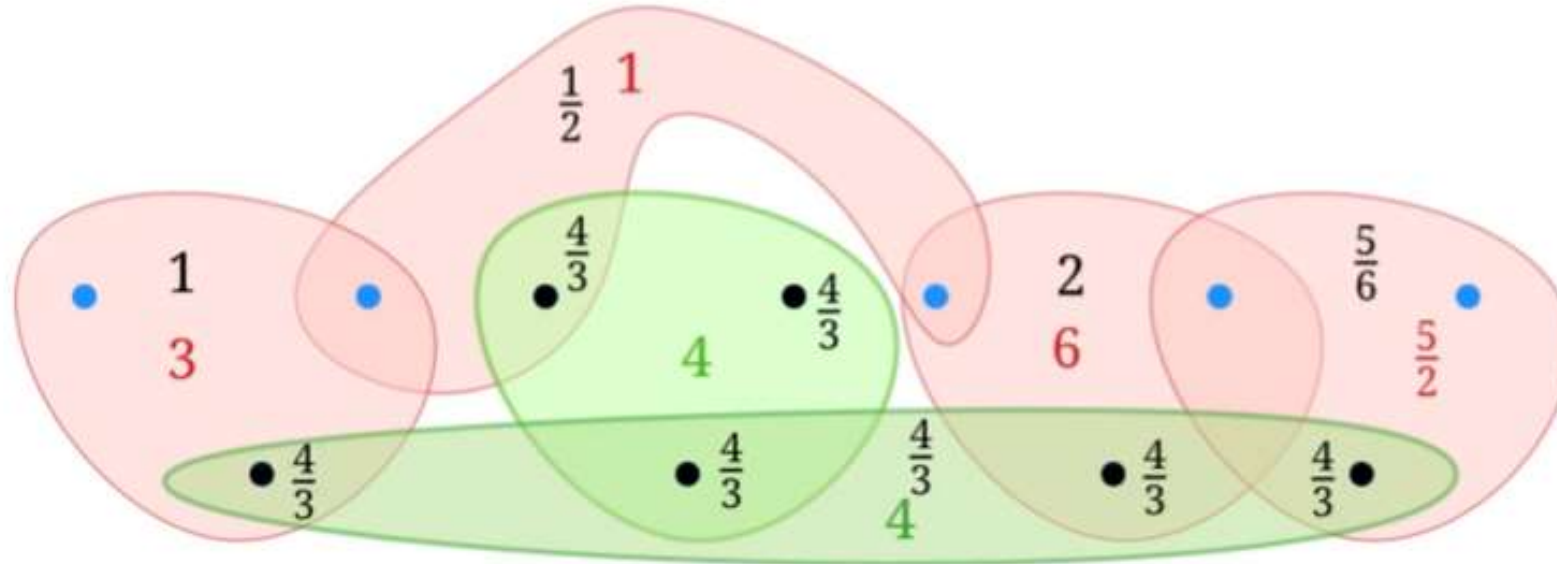
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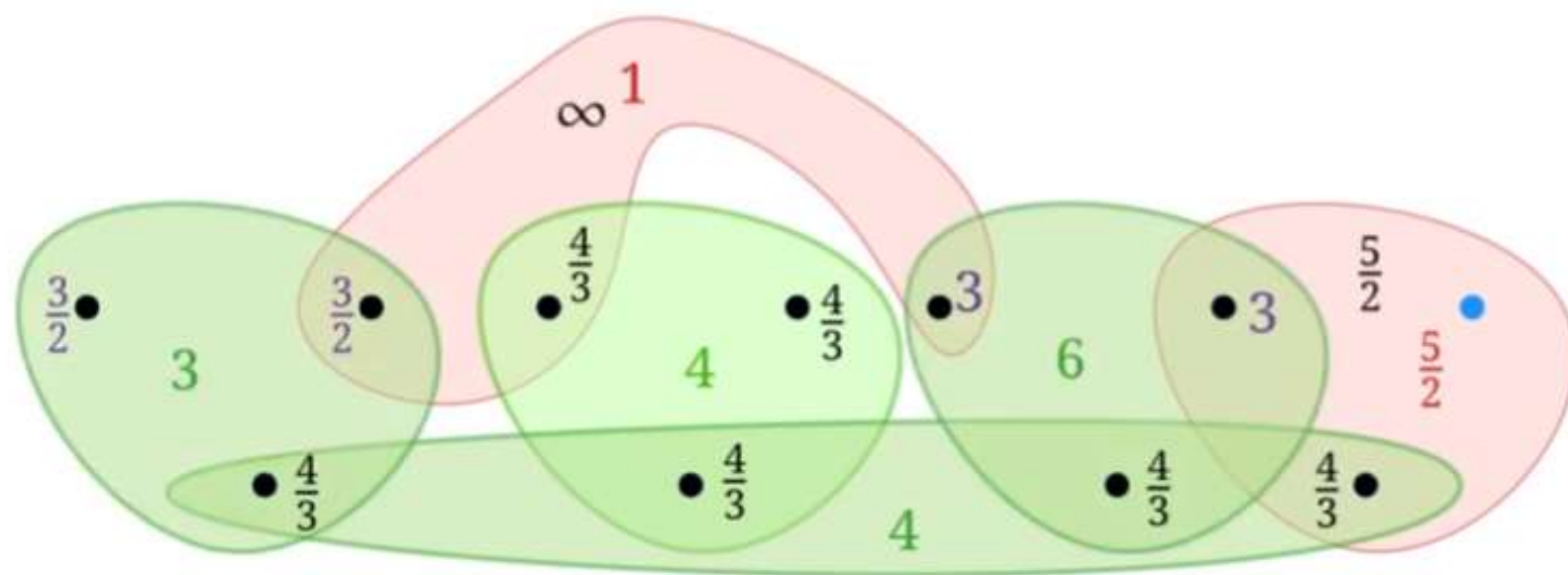
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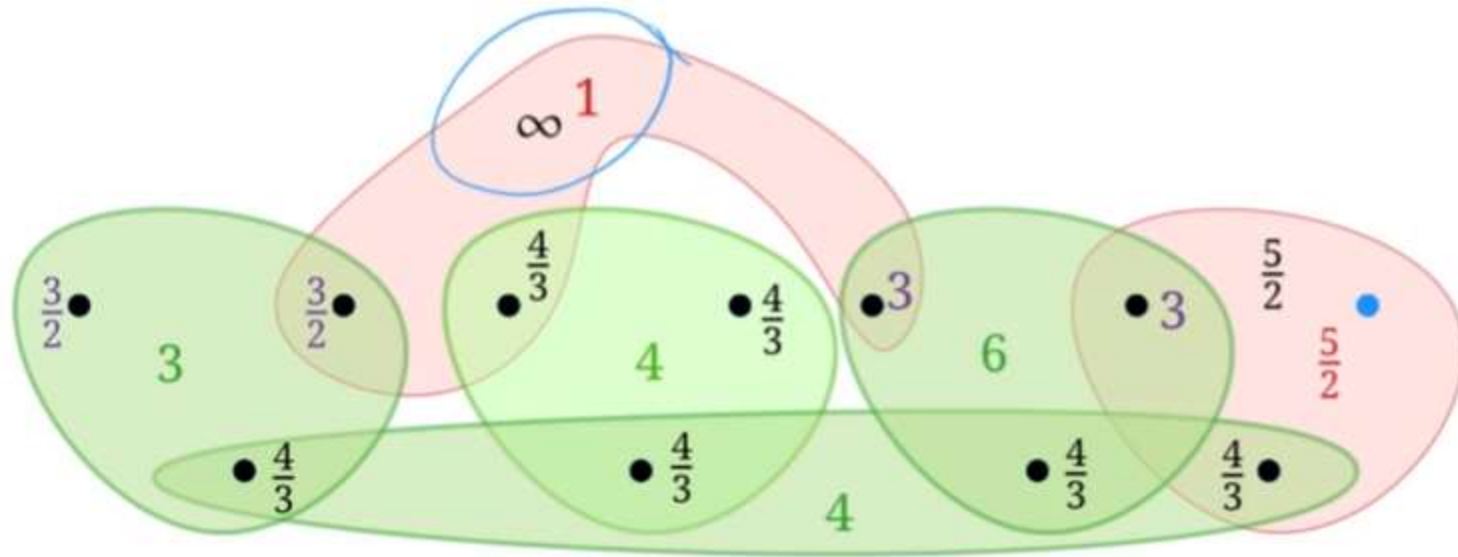
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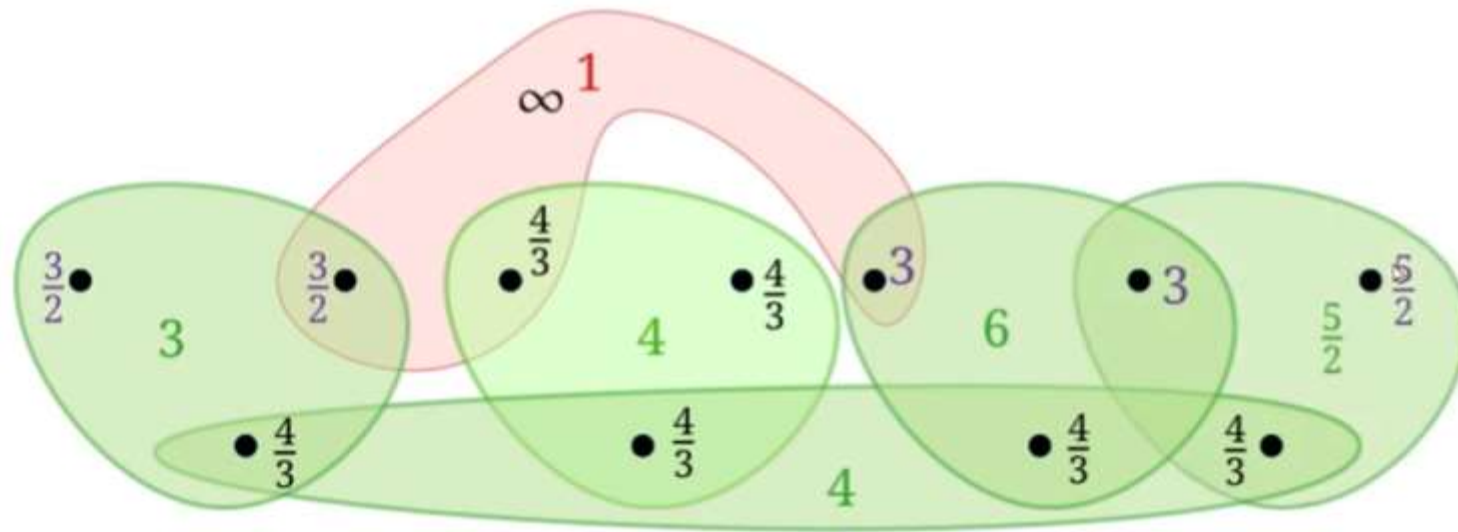
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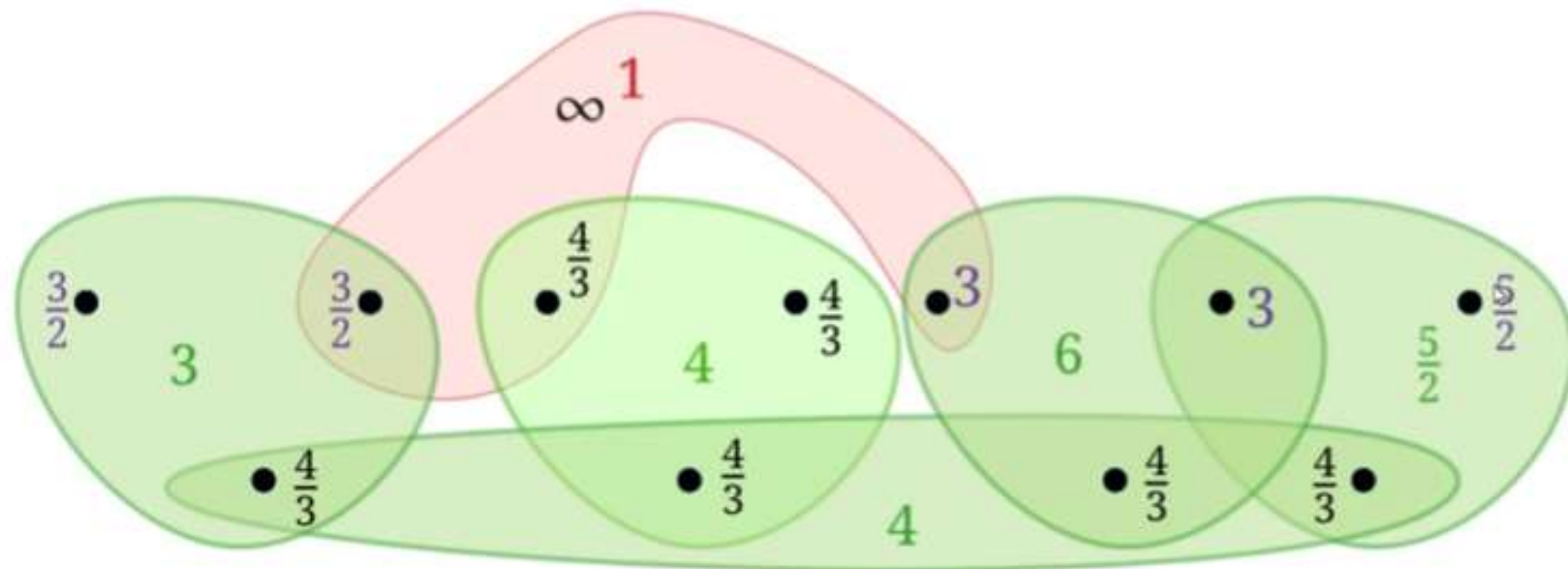
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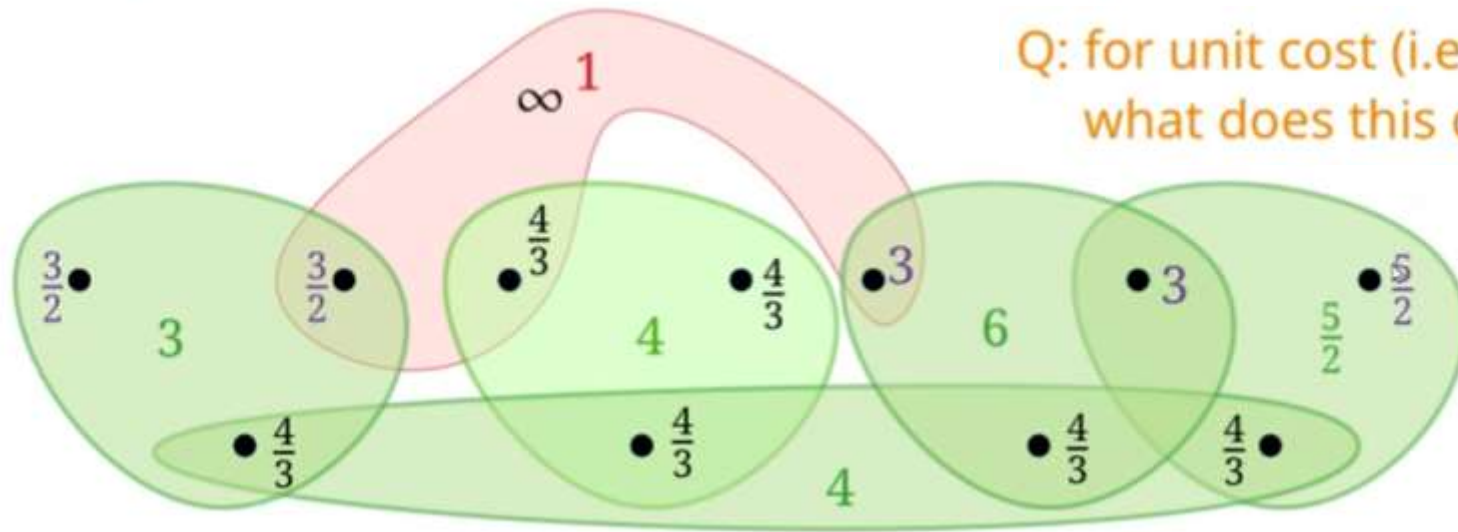
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$$\text{total cost: } \sum_{u \in U} \text{price}(u)$$

**Greedy:** Always choose the set with minimum per-element cost.



Q: for unit cost (i.e., cardinality set cover), what does this correspond to?

always pick the set that covers most of the remaining elements



# Greedy for SETCOVER

GreedySetCover( $U, S, c$ )

$C \leftarrow \emptyset$

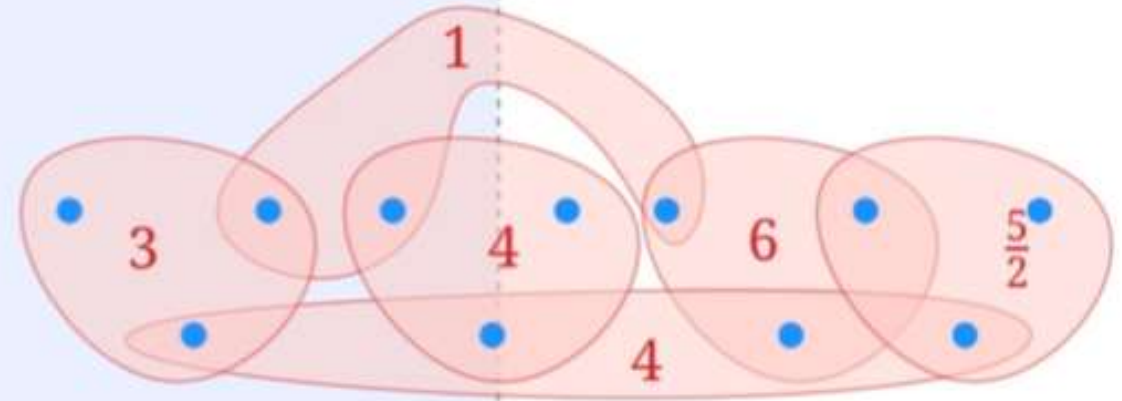
$S' \leftarrow \emptyset$

// covered elements

// picked sets

return  $S'$

// Cover of  $U$



# Greedy for SETCOVER

GreedySetCover( $U, S, c$ )

$C \leftarrow \emptyset$

$S' \leftarrow \emptyset$

while  $C \neq U$  do

$S \leftarrow$  set in  $S$  that minimizes  $\frac{c(S)}{|S \setminus C|}$

    foreach  $u \in S \setminus C$  do

        price( $u$ )  $\leftarrow \frac{c(S)}{|S \setminus C|}$

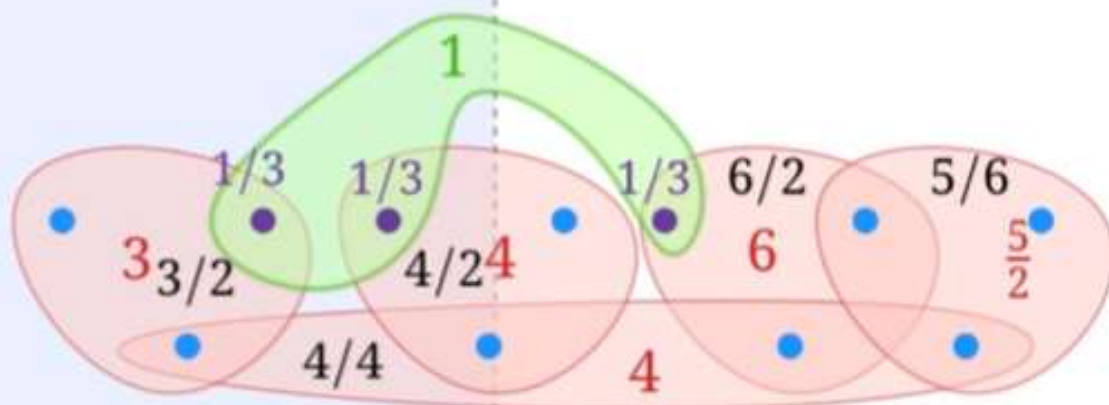
$C \leftarrow C \cup S$

$S' \leftarrow S' \cup \{S\}$

return  $S'$

// covered elements

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// Cover of  $U$

How does the algorithm continue?